The dataset first dropped the non-numeric and metadata (linguistic properties of artist and song) columns such as song name, artist, and date were removed, as they do not contribute to the audio feature analysis. Rows here the genre remained empty were also removed, giving 50000 songs in total. The train-test split was then performed before any preprocessing, dimensionality reduction, or modeling, ensuring there was no data leakage. Specifically, for each genre with at least 500 songs, 500 were randomly selected using a fixed seed (random_state=11034601) for the test set, and the remaining songs were assigned to the training set. This guarantees that the test data was never seen or used during training, transformation, or model tuning, preserving the integrity of the evaluation process. After splitting, the training set ends up having 45000 songs, whereas the test set has 5000.

To prepare the music dataset for classification, several data cleaning and transformation steps were applied. Categorical variables were encoded: the musical key was mapped from string labels (e.g., 'C#', 'G') to integers (0–11), and the mode (Major/Minor) was converted to binary values. Missing and invalid values were handled using imputation strategies. Specifically, missing values in audio features were filled using the mean of their respective columns, and negative values in the duration_ms column were imputed using the mean duration of valid songs. Additionally, non-numeric entries in the tempo column were coerced into numeric form, and any remaining invalid entries were either imputed or filtered. Finally, the dataset was scaled using StandardScaler before applying dimensionality reduction (PCA), ensuring consistent input for downstream clustering and classification models. This data preprocessing process was done for both training and test sets.

To reduce dimensionality and improve model efficiency, Principal Component Analysis (PCA) was applied to the standardized feature set. The features were standardized using StandardScaler, while categorical data (key and mode) were excluded from PCA operation. PCA retained 9 components, capturing approximately 96.67% of the total variance in the data. This lower-dimensional representation preserved the core structure of the feature space while eliminating redundancy. The lower dimension cluster is shown in Figure 2. The PCA visualization (Figure 2) shows that some genres like Classical and Hip-hop form distinct clusters, while others overlap heavily, suggesting that those overlapping genres share similar lower dimensional traits and are thus harder to tell apart. The acoustic features in the dataset are not normally distributed, but since the primary goal of PCA here is dimensionality reduction rather than statistical inference, this limitation was therefore left as-is. A silhouette analysis on PCA-reduced data revealed that the optimal number of clusters is 2 (Silhouette Score = 0.329), suggesting that while the original dataset includes 10 labeled genres, the underlying acoustic features form only two well-separated groupings, which helps explain the classification difficulty across some genres.

For the classification task, multiple models including random forest, XGBoost, MLP, were trained, examined, and their AUC on the test set were compared. Before training the model, the genres (y_train and y_test) were encoded into numeric values using LabelEncoder.

Multi-layer Perceptron (MLP) was selected as it achieved the highest AUC among all. This may be due to its flexibility in modeling complex, non-linear relationships between audio features and music genres. The MLP neural network model achieved a macro-average AUC of 0.924 and an overall accuracy of 56.14% on the test set. While the accuracy is fairly low, it is necessary to keep in mind that in multi-class classification, it is more likely for the model to be wrong than right, and thus even an accuracy of 56.14% is very meaningful. Performance varied by genre: the model performed well on Classical (F1 = 0.83) and Anime (F1 = 0.75), likely due to more distinct audio profiles. In contrast, genres like Hip-Hop (F1 = 0.37) and Alternative (F1 = 0.40) were harder to classify, potentially due to feature overlap with other genres (Figure 3). The ROC curve graph (Figure 4) further confirmed the difference across genres. The fairly balanced macro and weighted averages suggest no extreme imbalance in class-wise performance (Figure 3). This suggests that deep learning models, when combined with effective dimensionality reduction and clustering, can uncover the complex patterns behind genre classification.

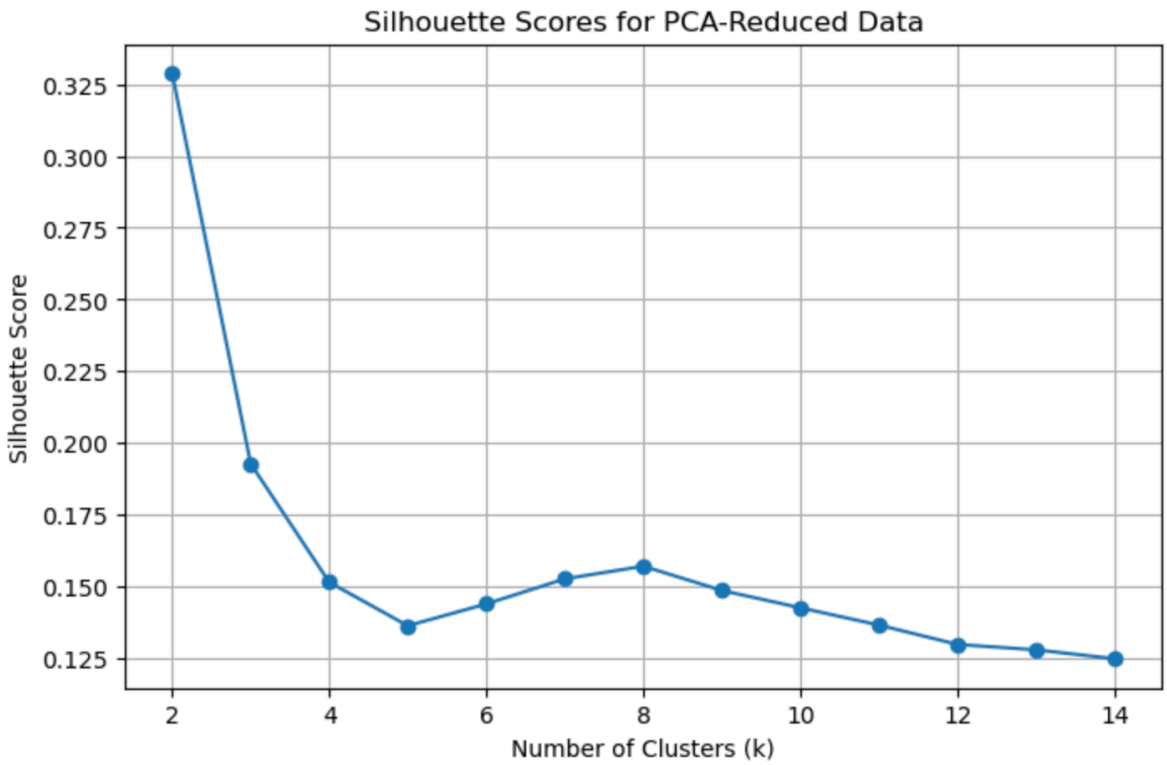
I believe the most important factor contributing to the model's classification success was the use of PCA to reduce noise and compress high-dimensional audio features into a more structured space, allowing the MLP to learn more distinct genre patterns. Additionally, preserving categorical features like key and mode without distorting them through scaling helped retain essential musical information relevant to genre. The combination of strong preprocessing, dimensionality reduction, and genre-aware feature engineering was key to the model achieving a high macro-AUC of 0.924.

Final AUC: 0.924.

Extra credit:

To examine whether speechiness affects popularity, a linear regression was performed between the two. The correlation was 0.2243, indicating a modest positive relationship. The regression showed that a unit increase in speechiness predicts a 34.39 point increase in popularity, though the $R^2$ was 0.0503, so speechiness explains about 5% of the variance (Figure 5). This is a non-trivial finding since speechiness isn't an obvious feature, yet it has a modest impact on popularity. Spoken genres like Hip-Hop may be more affected than other genres. This exploration highlights a subtle link between lyrics and popularity.

Figure 1: Silhouette scores for PCA-Reduced Data




```
Best number of clusters: 2 (Silhouette Score = 0.329)
```

Figure 2: Cluster visualization of the genres uncovered in the lower dimensional space

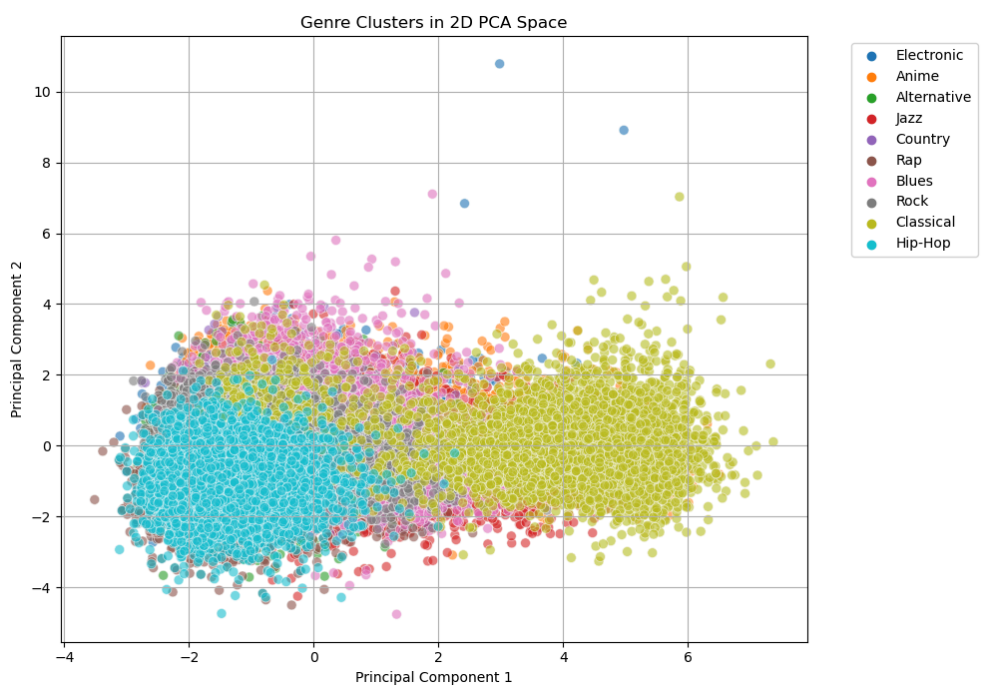


Figure 3: MLP classification report and AUC score on test set

```
MLP Accuracy: 0.5614
MLP Macro-AUC: 0.92355

Classification Report:
               precision    recall  f1-score   support

 Alternative       0.40      0.40      0.40       500
       Anime       0.76      0.75      0.75       500
       Blues       0.59      0.49      0.53       500
   Classical       0.84      0.82      0.83       500
     Country       0.57      0.51      0.54       500
  Electronic       0.62      0.64      0.63       500
     Hip-Hop       0.44      0.32      0.37       500
        Jazz       0.52      0.51      0.52       500
         Rap       0.44      0.51      0.47       500
        Rock       0.48      0.66      0.56       500

    accuracy                           0.56      5000
   macro avg       0.56      0.56      0.56      5000
weighted avg       0.56      0.56      0.56      5000
```

Figure 4: Graph of MLP ROC Curves by Genre

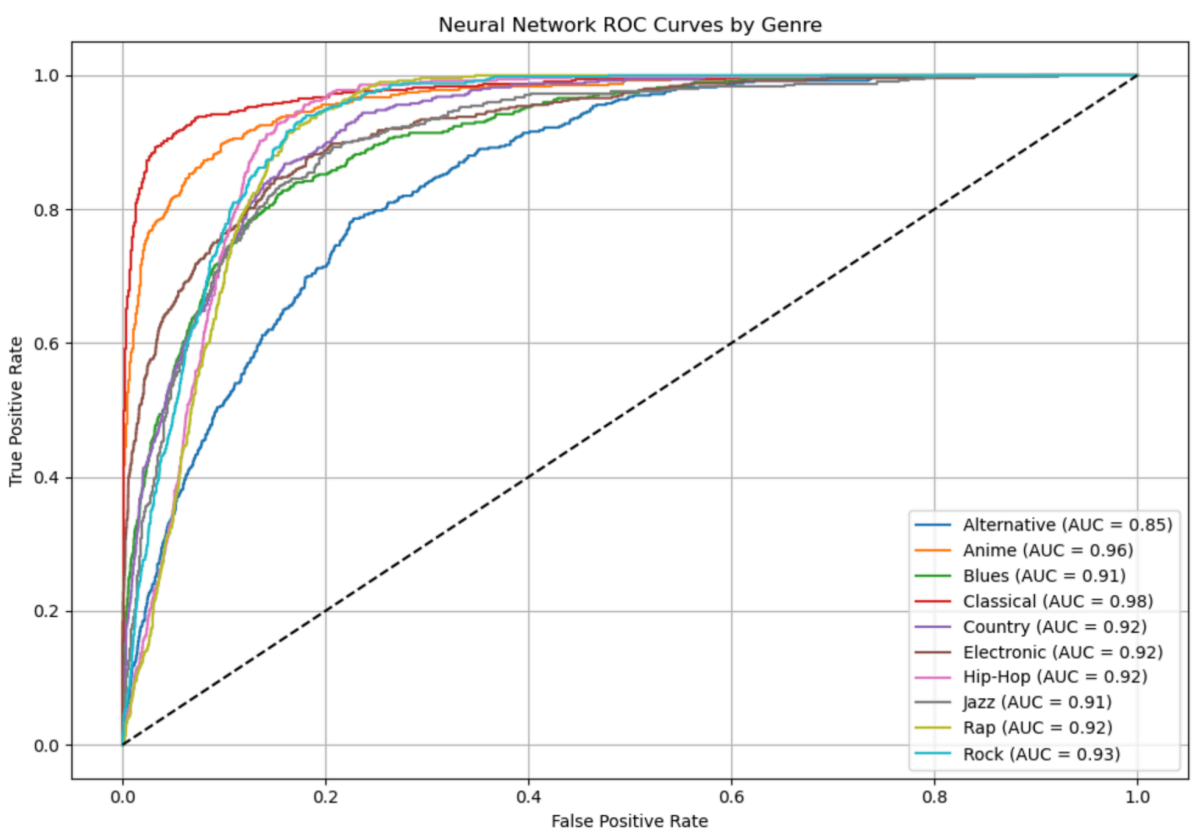
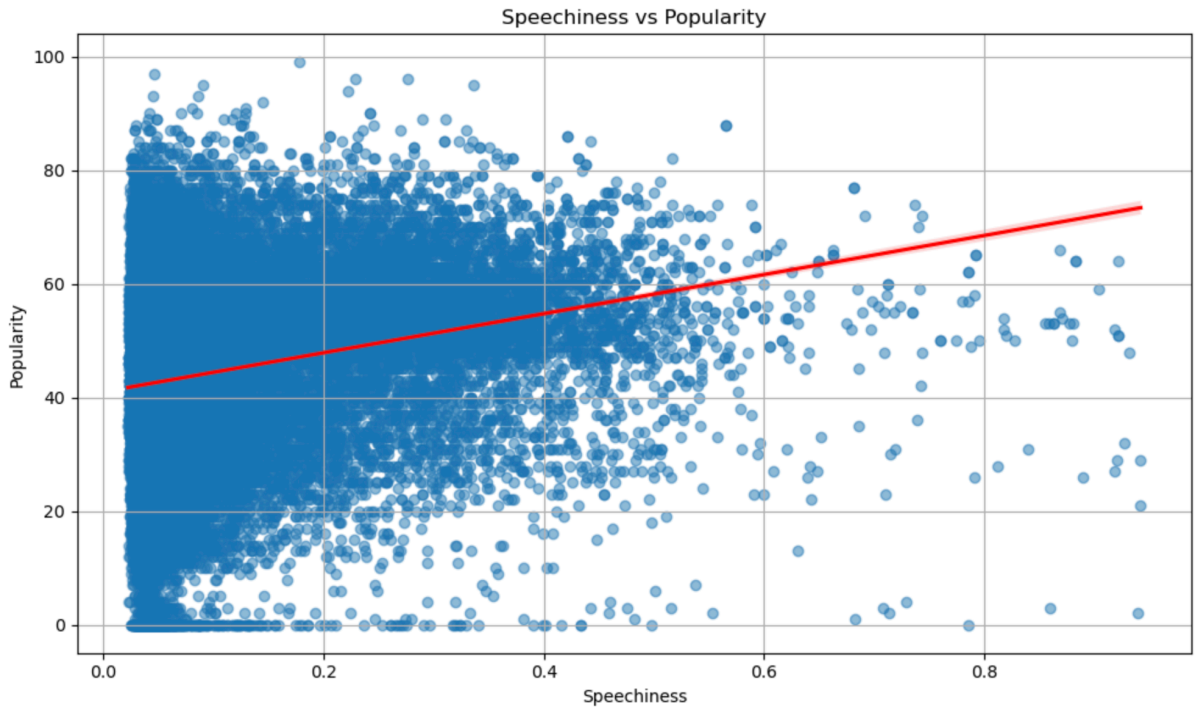
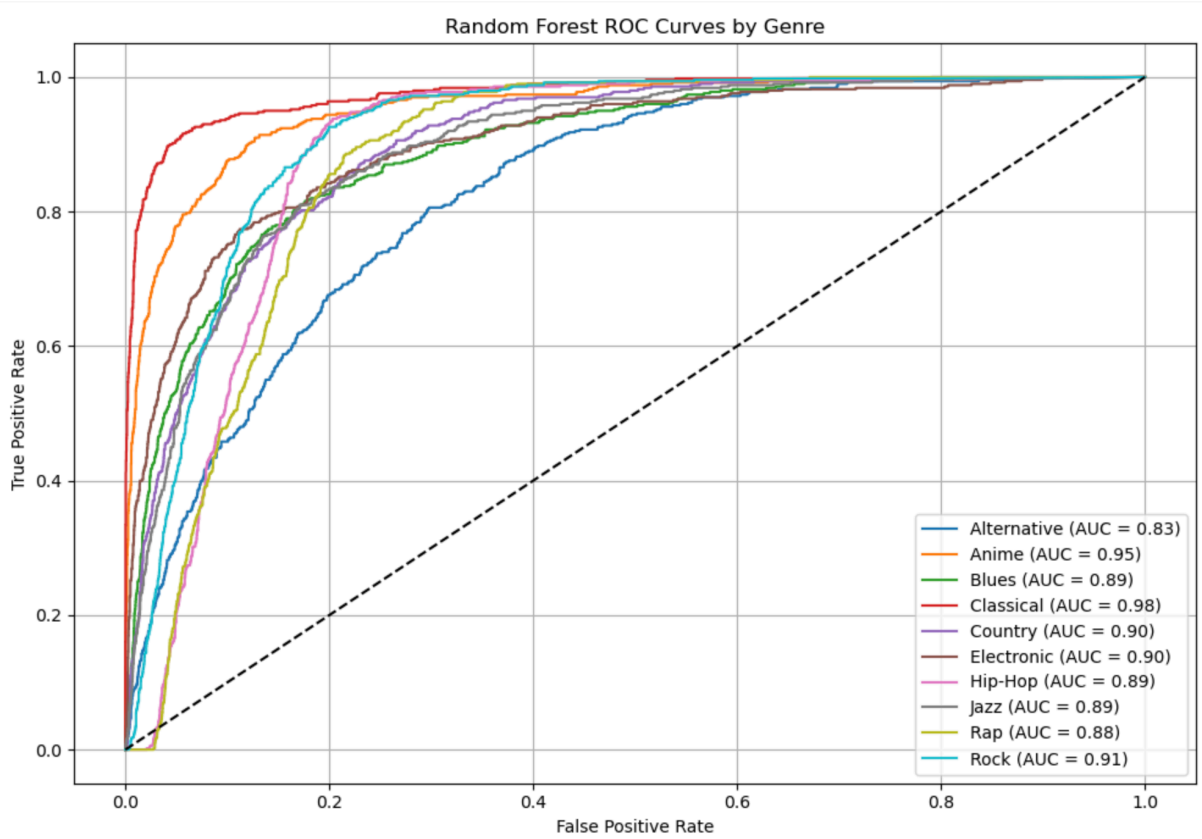
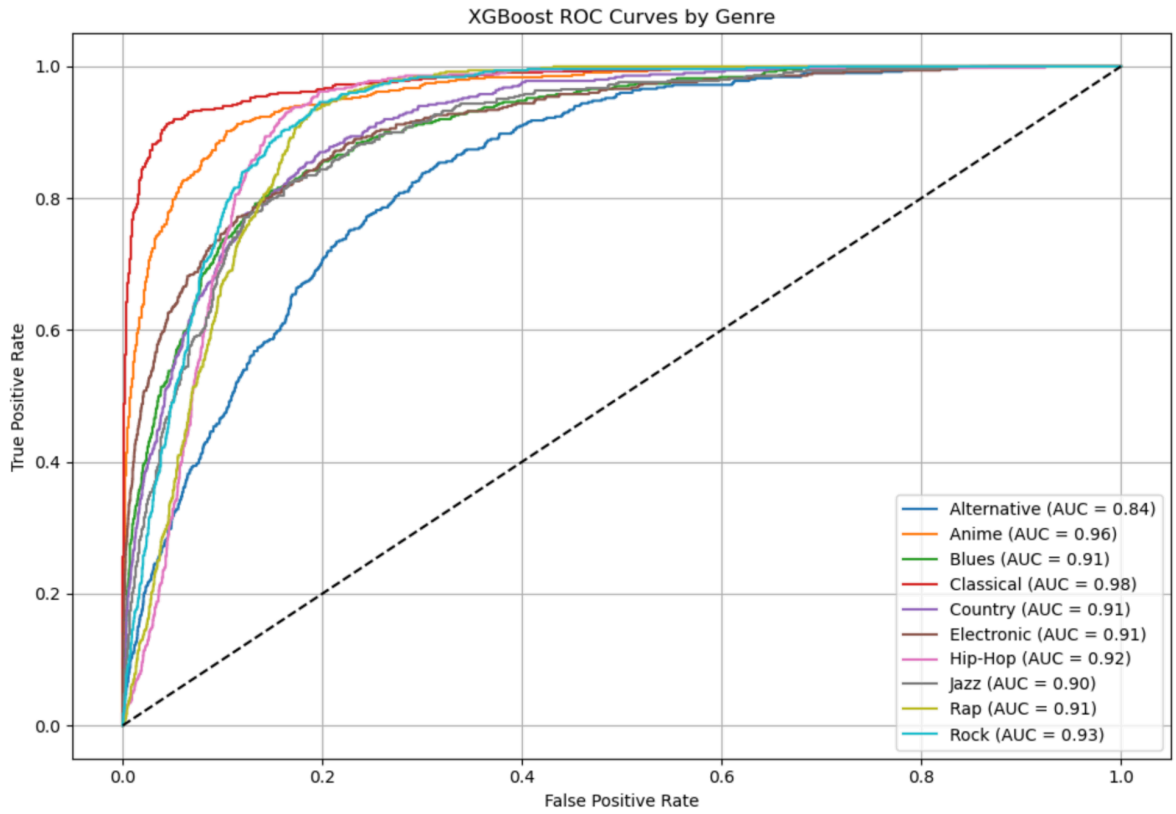


Figure 5: Scatter plot of Speechiness vs Popularity




ROC Curves from other models (Random Forest, XGBoost)




XGBoost ROC Curves by Genre
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Alternative (AUC = 0.84)
Anime (AUC = 0.96)
Blues (AUC = 0.91)
Classical (AUC = 0.98)
Country (AUC = 0.91)
Electronic (AUC = 0.91)
Hip-Hop (AUC = 0.92)
Jazz (AUC = 0.90)
Rap (AUC = 0.91)
Rock (AUC = 0.93)